

Bitcoin Market Sentiment & Trader Data Analysis

This report combines Bitcoin market sentiment data (Fear/Greed Index) with trading data from Hyperliquid. The goal is to study how trader performance connects with market psychology, find patterns, and share useful insights.

1. Data Import & Merging

We first load two datasets:

- **Bitcoin Market Sentiment Dataset** – Fear/Greed index values for each day.
- **Trader Data** – trade details such as execution price, size, side, profit/loss (PnL), and fees.

After cleaning and formatting the dates, we merged both datasets on the date column. This gave us a single dataset with both **sentiment** and **trader metrics** for further analysis.

2. Exploratory Data Analysis (EDA)

In this step, we checked:

- Shape, columns, and data types.
- Missing values and duplicates.
- Summary statistics of key fields (execution price, trade size, and closed PnL).

This helped us understand the data quality and basic distributions.

3. Aggregated Analyses

We created daily summaries to study activity levels. Metrics include:

- Number of trades per day.
- Total closed PnL per day.
- Average trade size per day.
- Average Fear/Greed index per day.

These summaries allowed us to study how trading performance changes over time.

4. Trend Analysis Over Time

We studied the following trends:

- **Number of trades per day** – to see activity changes.
- **Daily average PnL** – to see profit/loss trends.
- **Fear/Greed index movement** – to compare trader outcomes with market psychology.

This shows how trading behavior changes during fear vs greed phases.

5. Relationship Between Sentiment & Trader Metrics

We compared sentiment with trading outcomes:

- **PnL across sentiment phases** – to see if profits are higher in Fear or Greed.
 - **Trade size vs sentiment** – to see if traders risk more in Greed.
 - **Correlation between sentiment score and PnL** – to measure the strength of the link.
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6. Interactive Visualizations & Insights

Using Plotly, we built interactive charts:

- Daily sentiment and PnL trends.
- Trade activity vs Fear/Greed index.
- Bar charts of average PnL in different sentiment phases.

These visualizations make it easier to explore patterns in the data.

7. Key Insights

- Trade activity is higher when sentiment is strong (either Fear or Greed).
 - PnL moves in line with sentiment cycles.
 - In many cases, traders show higher profits in Greed phases, suggesting they take more risks when the market is positive.
 - Trade sizes also change slightly with sentiment, showing different levels of risk appetite.
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8. Conclusion

This assessment:

- Successfully merged sentiment and trading datasets.
- Performed daily trend analysis of trades and PnL.
- Studied relationships between sentiment and trader performance.
- Created interactive dashboards to explore the results.

Takeaway:

Market sentiment strongly influences trading outcomes. Greed often leads to higher profits but also higher risks. Fear periods usually bring lower activity and more cautious behavior. These insights can support better trading strategies and risk management.